



Building RAG Agents with LLMs

Product Building with Agentic Coding

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.

Today, You Will...

Product Building with Agentic Coding

- Reuse an existing Skill
- Build a public-data Skill
- Create and verify a useful page
- Connect MCP, deploy, and update



Today's Route

Understand → choose → practice → apply

1

Understand

Coding models + their limits

2

Choose

One coding workspace

3

Practice

Reuse · build · verify ·
deploy

4

Apply

A URL + one verified update



Building RAG Agents with LLMs

Product Building with Agentic Coding

Code Is Also a Sequence of Tokens

A tiny completion—with a checkable result

```
def add(a, b):  
    return a + b
```

```
add(2, 3)
```

5



Run the example

Producing plausible code and passing the check are different.

Code-Focused Model Families

Examples already in this course: Qwen3-Coder and DeepSeek-Coder-V2

QWEN3-CODER

Qwen3-Coder

updated Dec 31, 2025

Qwen3 Coder WebDev 🌐

Generate HTML or React code from your web app description

🔥 ❤️ 1.15k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct

📄 Text Generation • 🗄️ 480B • Updated Aug 21, 2025 • ⬇️ 30.3k • ⚡ • ❤️ 1.37k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct-FP8

DEEPSEEK-CODER-V2

DeepSeekCoder-V2

updated Nov 27, 2025

🌟 deepseek-ai/DeepSeek-Coder-V2-Instruct

📄 Text Generation • 🗄️ 236B • Updated Aug 21, 2024 • ⬇️ 9.65k • ❤️ 700

🌟 deepseek-ai/DeepSeek-Coder-V2-Base

📄 Text Generation • 🗄️ 236B • Updated Jul 3, 2024 • ⬇️ 2.83k • ❤️ 82

🌟 deepseek-ai/DeepSeek-Coder-V2-Lite-Base

📄 Text Generation • 🗄️ 16B • Updated Jul 3, 2024 • ⬇️ 8.36k • ❤️ 118

Code-Model Ecosystem

Model family overview · not a current leaderboard

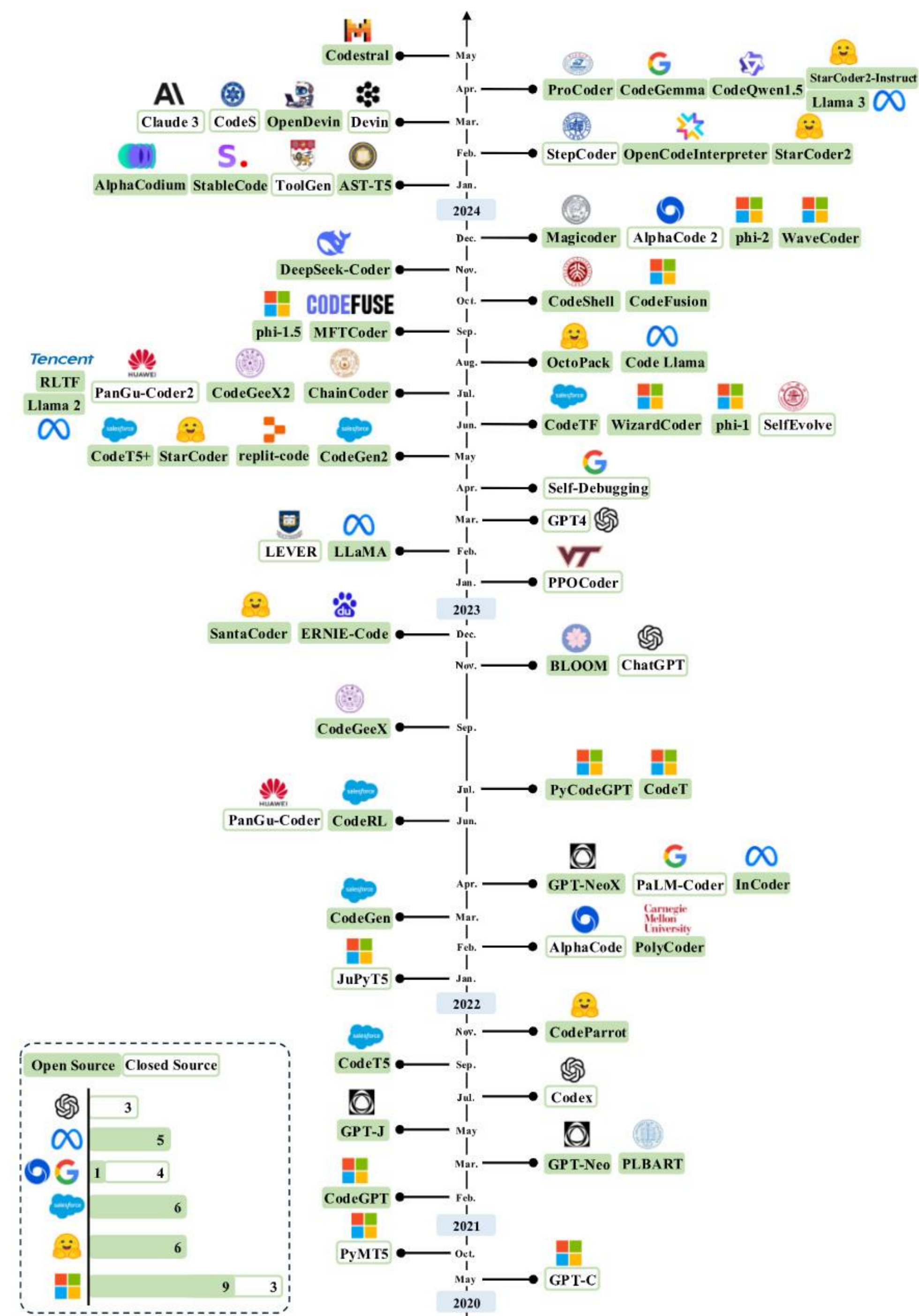
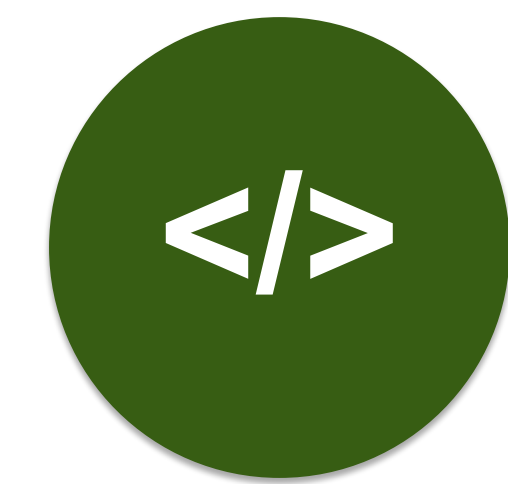


Fig. 1. A chronological overview of LLMs for code generation in recent years. The timeline was established mainly according to the release date. The models with publicly available model checkpoints are highlighted in green color.

Many families.
Different sizes.
Different
trade-offs.

Current Models, Concrete Tasks

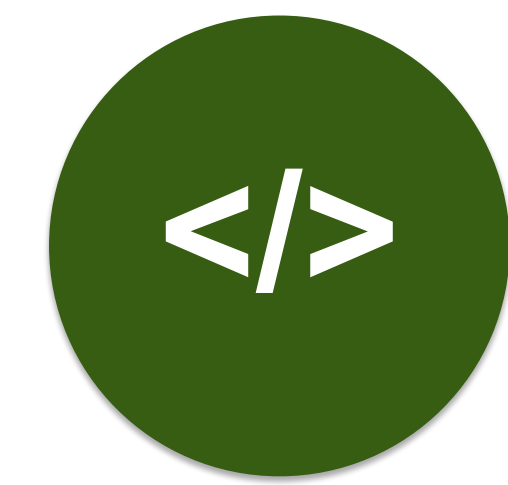
Representative choices · September 2026



GPT-6 Astra



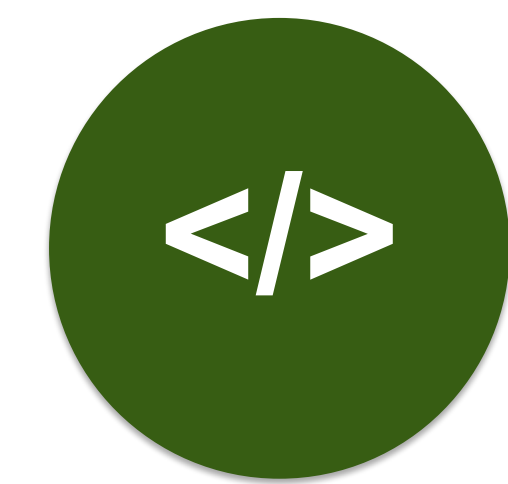
Implement a multi-file feature



Claude Fable 5.1



Trace a bug through a repository



Gemini 3.8 Flash



Iterate on a UI change

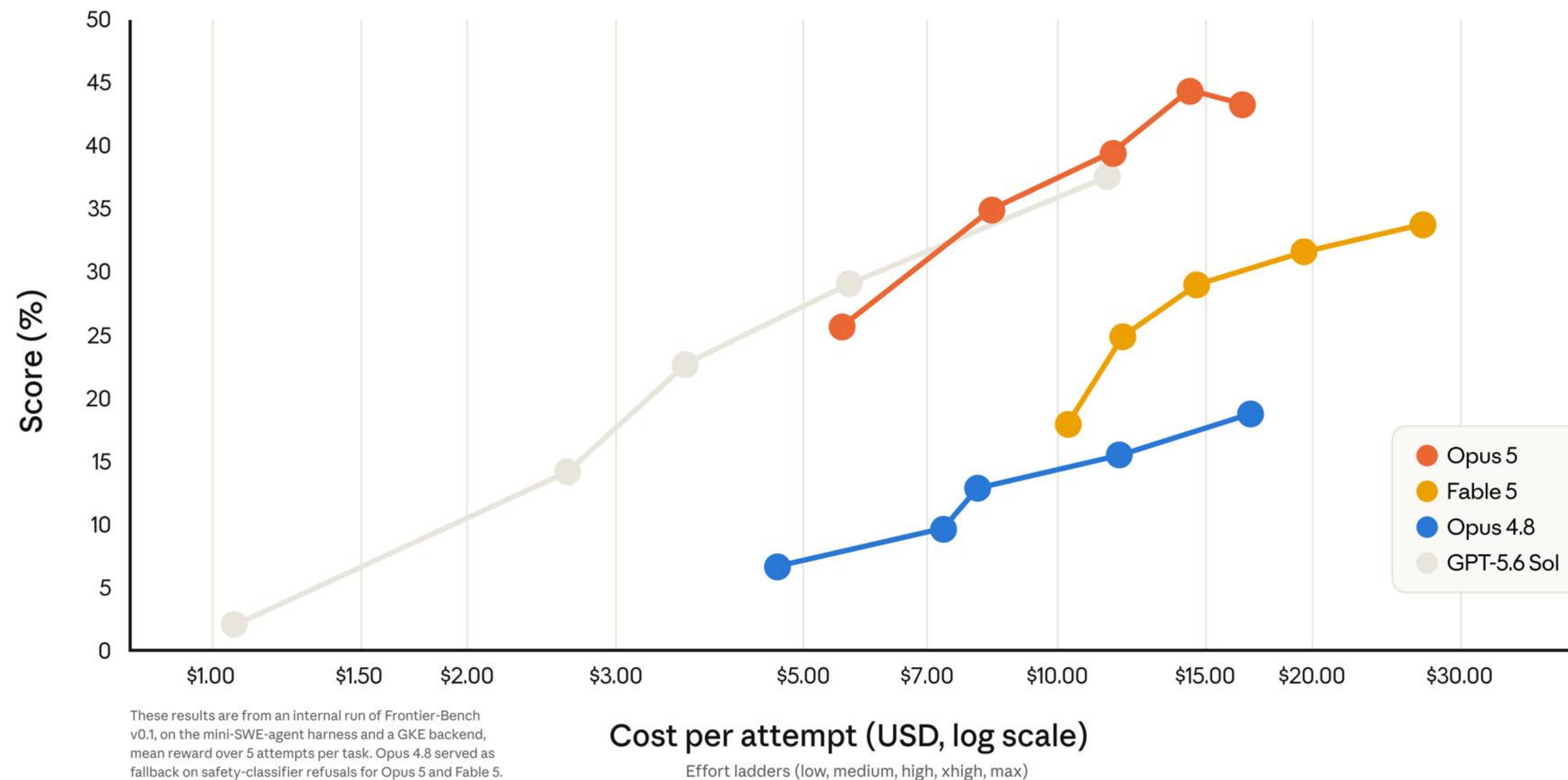
Example tasks—not exclusive capabilities or a ranking.

More Capability Can Cost More

Read both axes: task score and cost per attempt

Agentic coding by effort level

Frontier-Bench v0.1



What Makes It an Agent?

A model inside a tool-using feedback loop



The workflow—not just the model—determines the result.



Building RAG Agents with LLMs

Product Building with Agentic Coding

Vibe Coding vs. Agentic Coding

How you iterate versus how the work is executed

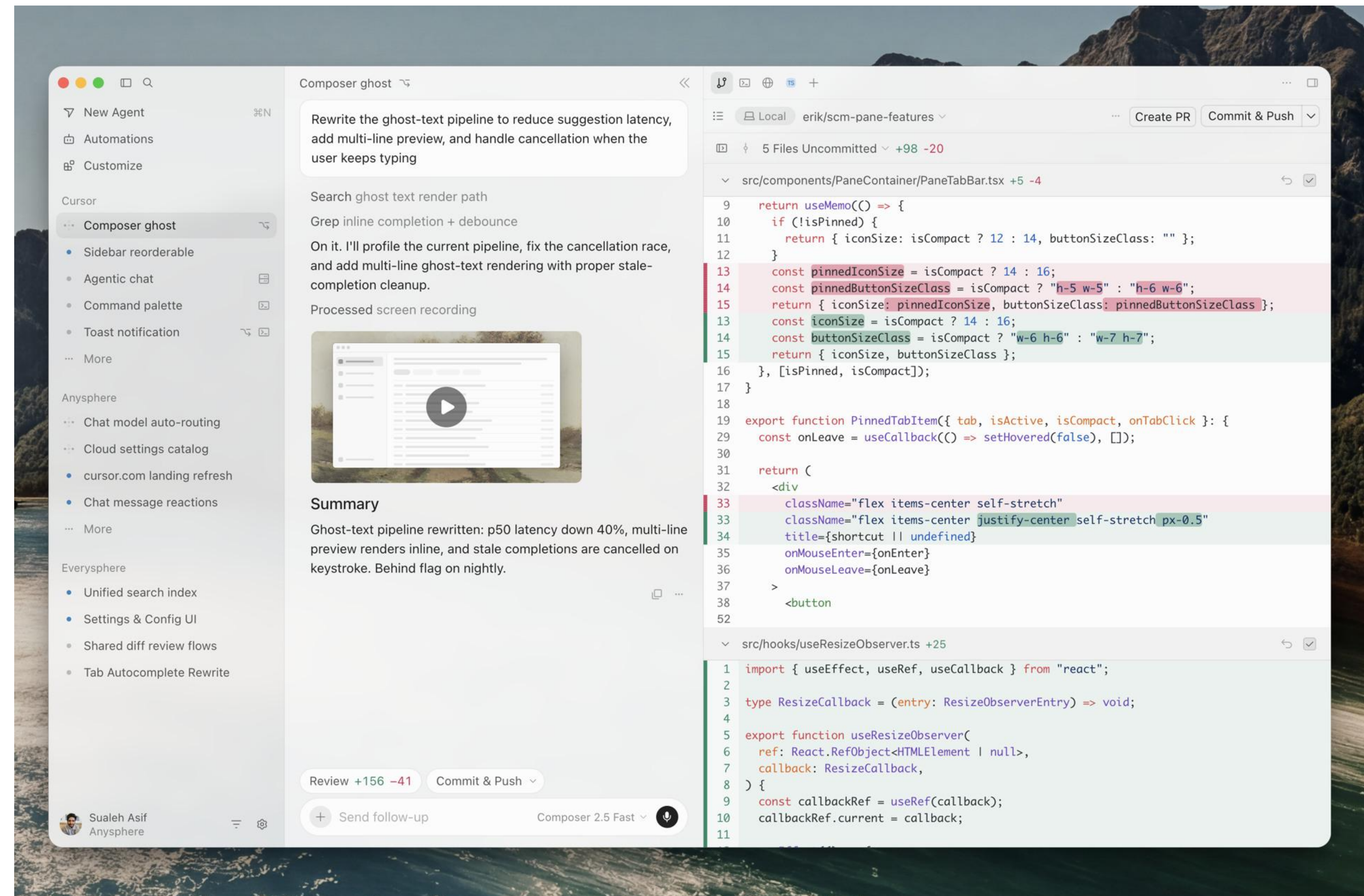
“Make it feel better.”

Intent → visual feedback

Vibe: quick iteration

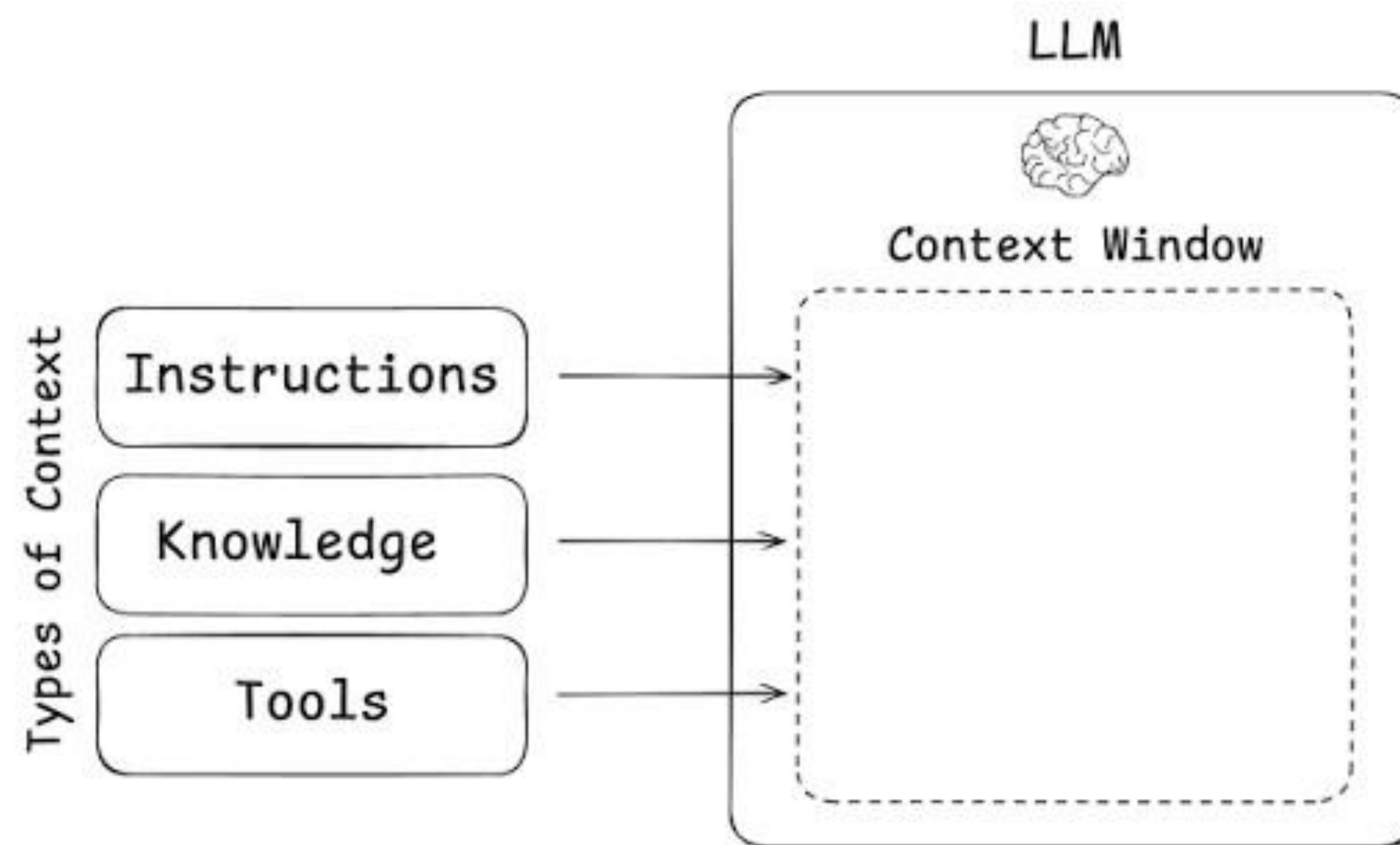
Agent: tool-based work

They can overlap. You still own the acceptance criteria.



Why Vibe Coding Fails

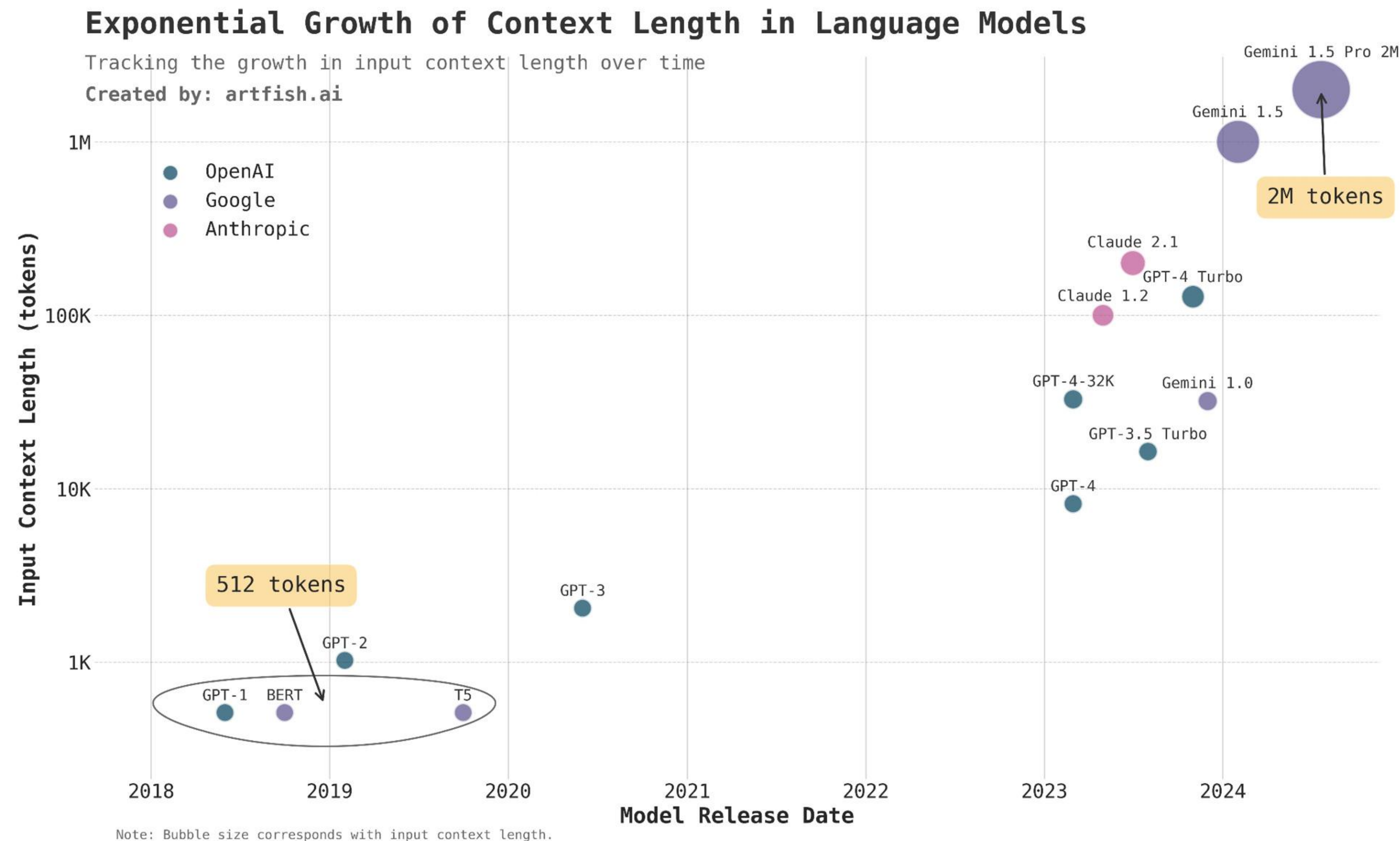
Instructions, context, and verification



- 1 Ambiguous instructions**
- 2 Missing or overloaded context**
- 3 Incorrect model output**

Context Windows Got Much Larger

Historical growth—not a promise of perfect recall



512

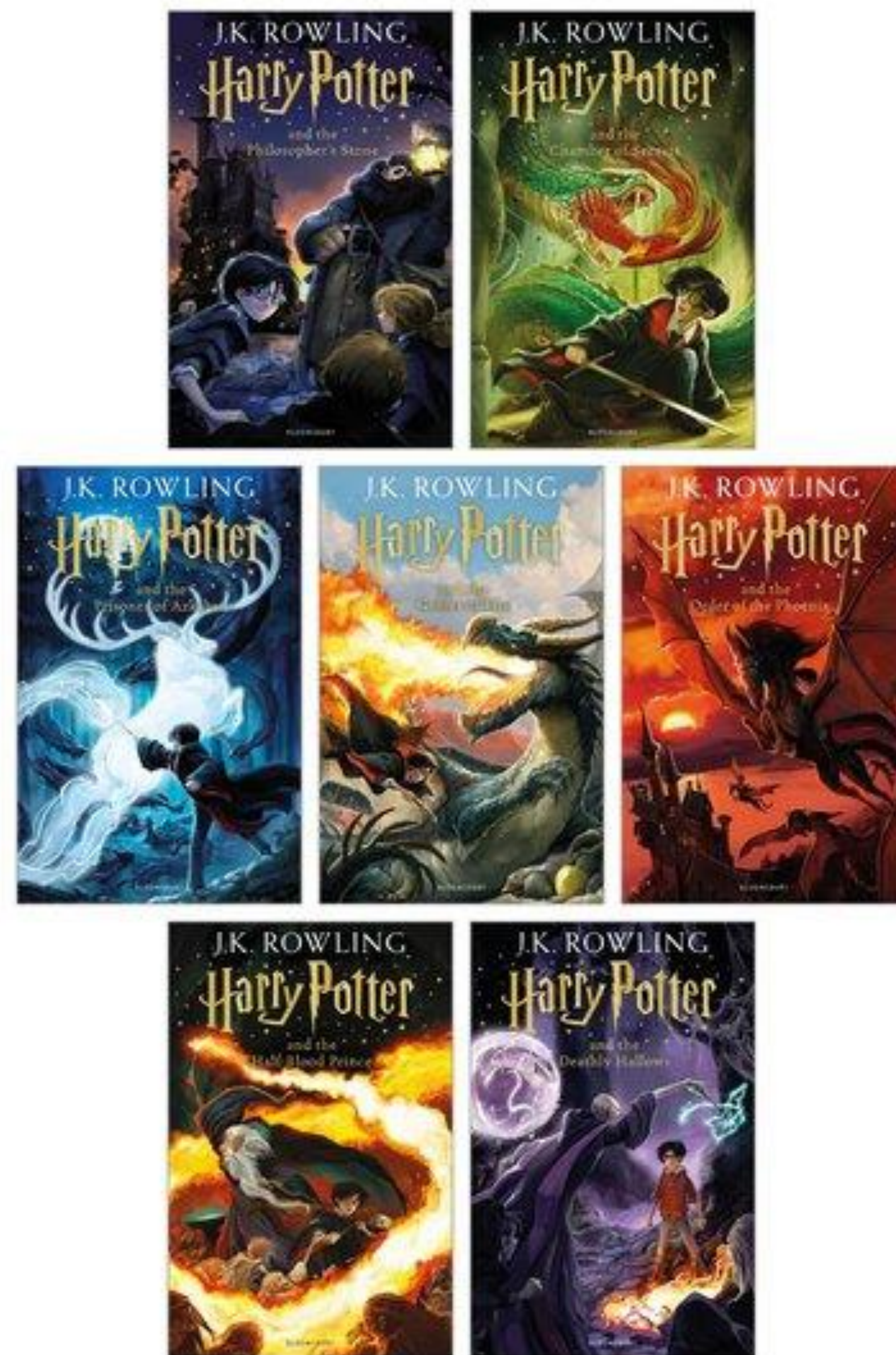


2M

tokens

How Much Is 1.5 Million Tokens?

The Harry Potter example



7 books

~1.5M

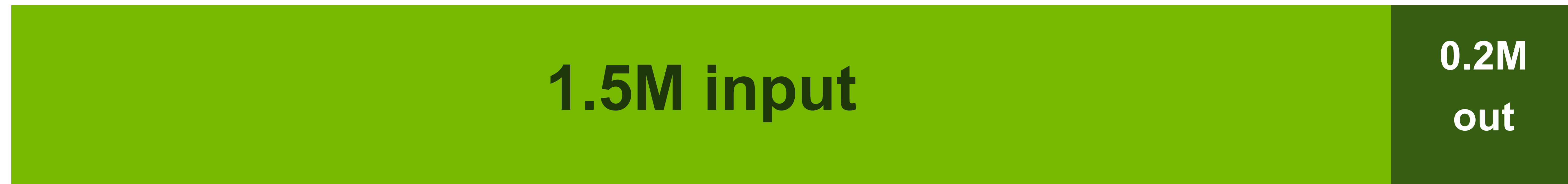
tokens

A scale estimate: token counts depend on edition and tokenizer.

A Large Context Still Has a Limit

Input and generated output must fit the model's budget

Harry Potter thought experiment



Astra context window: 1.05M

This does not fit in one Astra request. Split or retrieve.

Long Context Also Has a Price

An illustrative total across multiple calls



1.5M total input × \$20 / M **\$30**

0.2M total output × \$75 / M **\$15**

Illustrative total **\$45**

Standard, uncached, long-context rates

Example: 3 calls, each with 0.5M input and ~0.067M output.

Pack Context for the Next Decision

A working folder is more useful than a giant chat history

Brief

Relevant files

Failing case



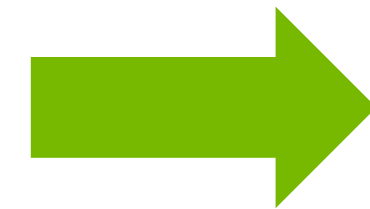
Goal: explore heritage
Keep: verified API facts
Failure: stale search
results
Next: fix the filter

Give the agent the smallest context that explains the next change.

Replace “Make It Better” with a Test

A concrete request changes the conversation

“Improve the app.”

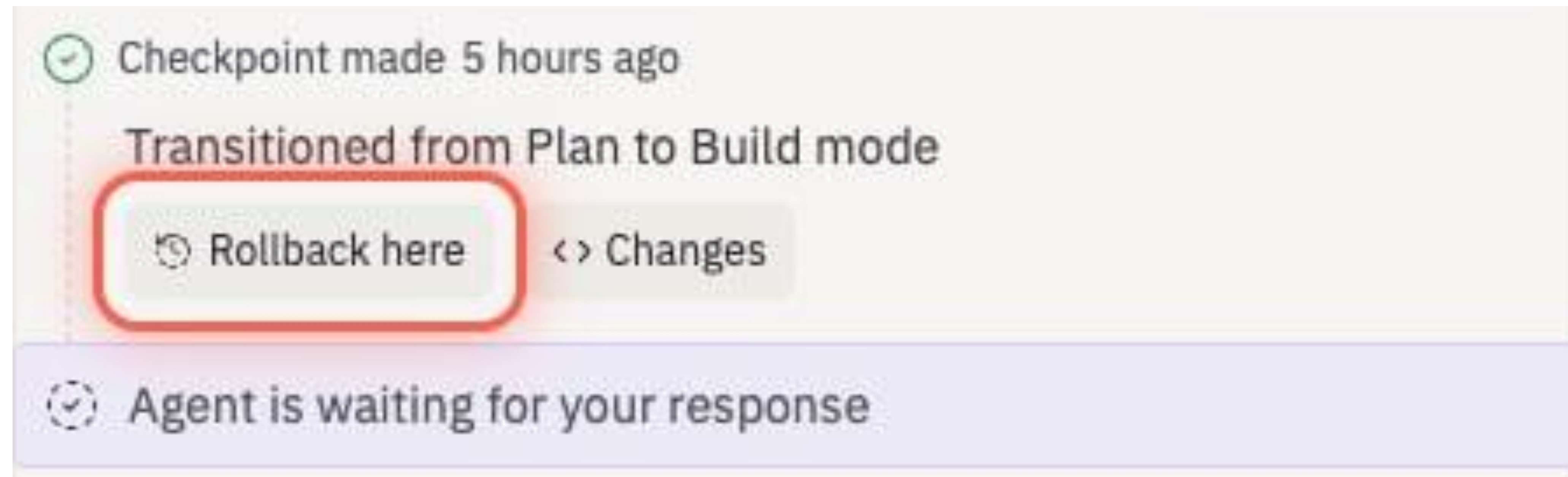


- 1 Open the page
- 2 Search for one item
- 3 Only matching items remain

An observable outcome beats a long list of adjectives.

Keep a Way Back

Checkpoint before a larger change



Save

Inspect

Restore

Use sample data. Scope permissions. Review before publishing.



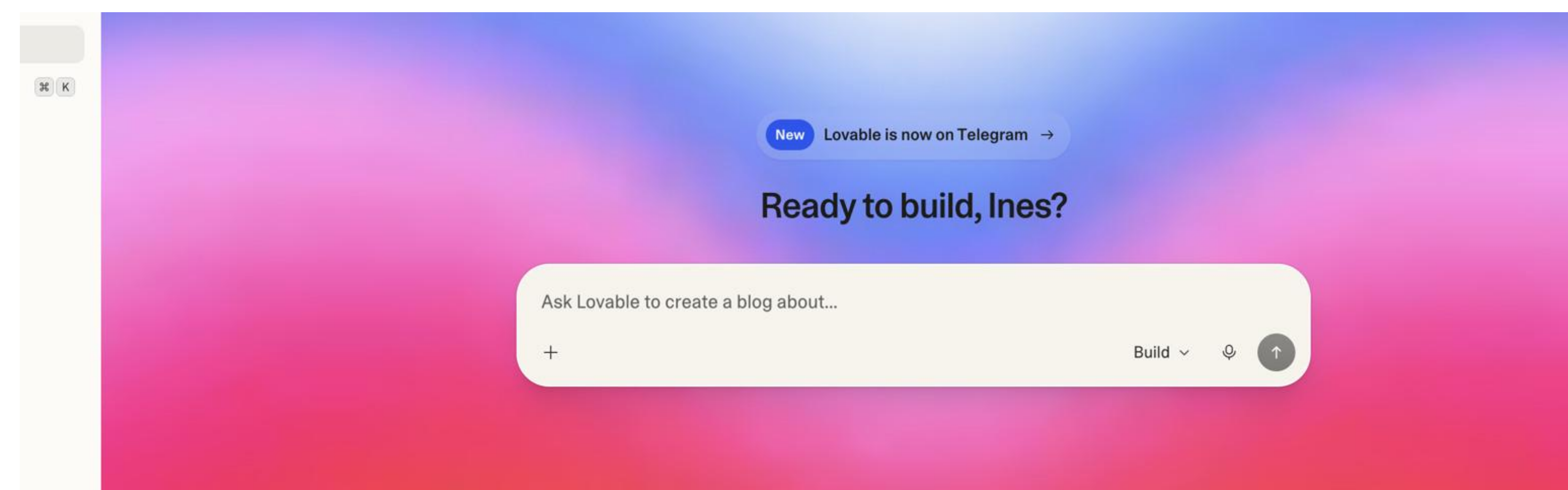
Building RAG Agents with LLMs

Product Building with Agentic Coding

Choose Your Starting Point

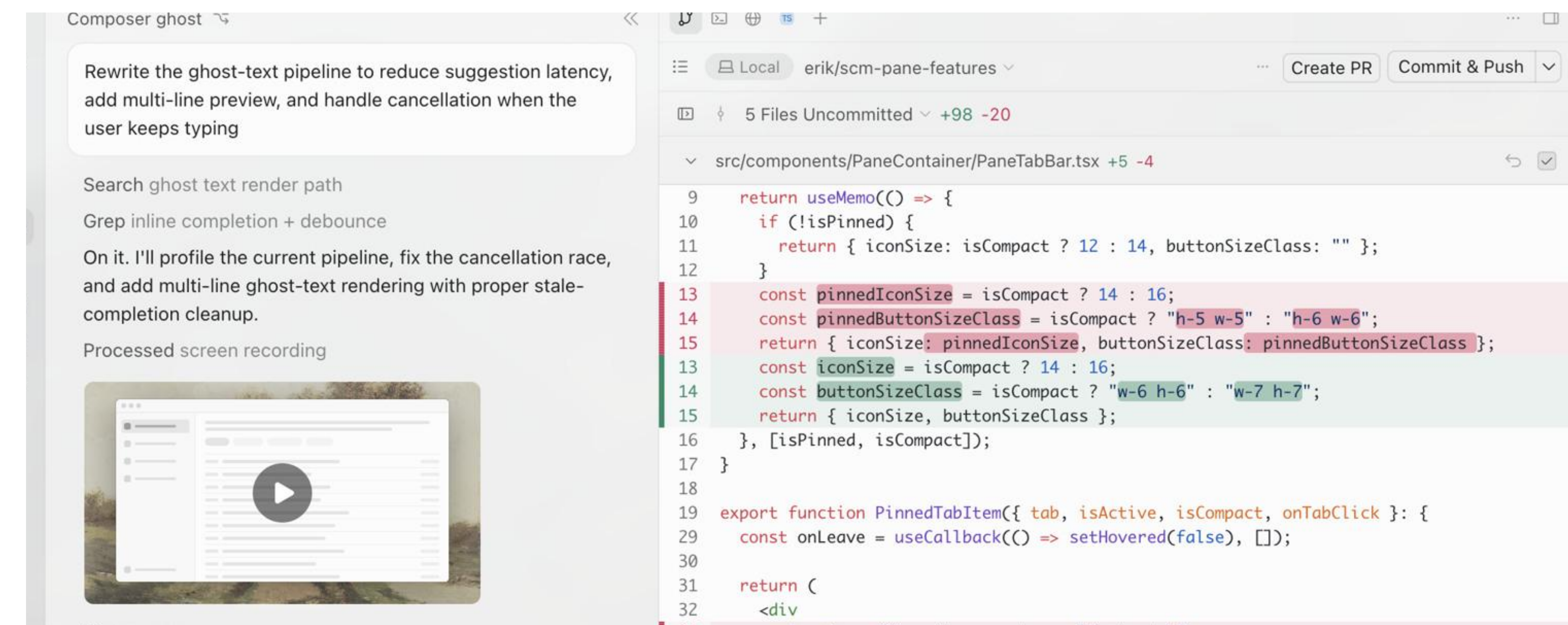
An idea-first builder or a repository-first agent

IDEA → PREVIEW



Lovable · Replit

FILES → PATCH

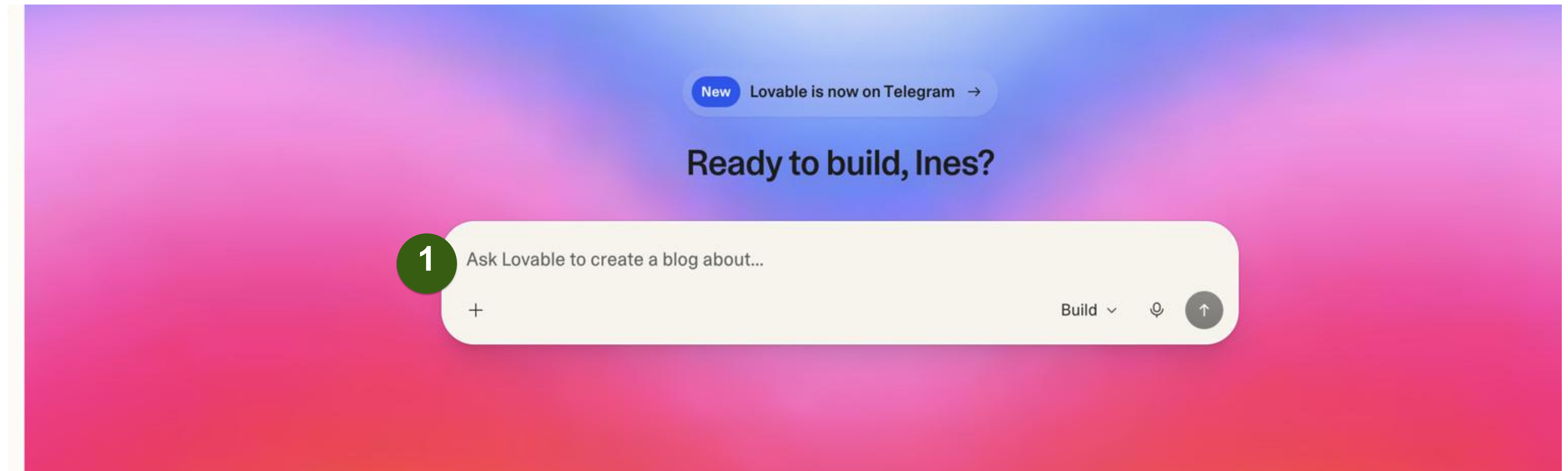


Cursor · Claude Code · Codex

Builders: demo · ChatGPT desktop Work: follow-along

Lovable: Describe the App

Watch where the first instruction goes



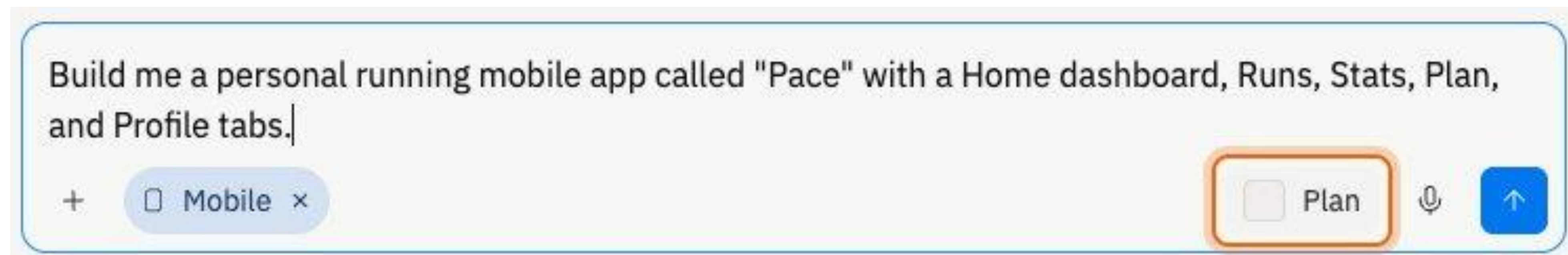
1 Describe the user, outcome, and constraints.

[Open ↗](#)

Replit: Plan Before Build

Read the plan before starting implementation

WRITE THE REQUEST

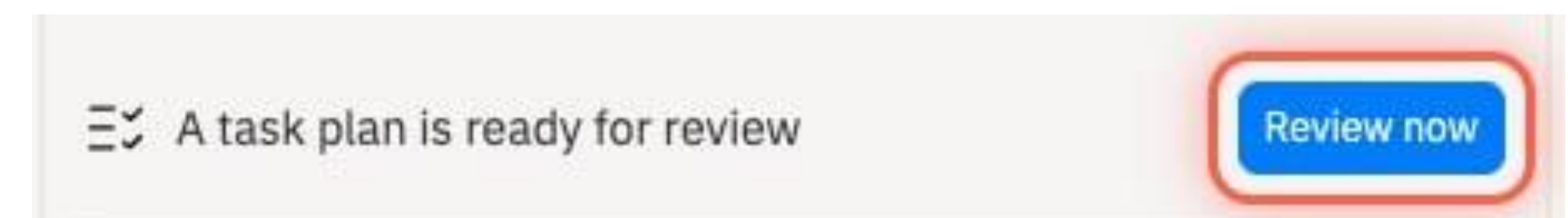


Build me a personal running mobile app called "Pace" with a Home dashboard, Runs, Stats, Plan, and Profile tabs.

+ Mobile x

Plan

REVIEW THE PLAN



A task plan is ready for review

Review now

Turn on Plan

Review now

Check scope, completion criteria, and what stays unchanged.

Cursor: Request → Diff

Check whether the patch matches the request

REQUEST

Rewrite the ghost-text pipeline to reduce suggestion latency, add multi-line preview, and handle cancellation when the user keeps typing

CODE DIFF



```
9 return useMemo(() => {
10   if (!isPinned) {
11     return { iconSize: isCompact ? 12 : 14, buttonSizeClass: '' };
12   }
13   const pinnedIconSize = isCompact ? 14 : 16;
14   const pinnedButtonSizeClass = isCompact ? 'h-5 w-5' : 'h-6 w-6';
15   return { iconSize: pinnedIconSize, buttonSizeClass: pinnedButtonSizeClass };
13   const iconSize = isCompact ? 14 : 16;
14   const buttonSizeClass = isCompact ? 'w-6 h-6' : 'w-7 h-7';
15   return { iconSize, buttonSizeClass };
16 }, [isPinned, isCompact]);
```

Inspect changed lines. Ask for a smaller patch if the scope drifts.

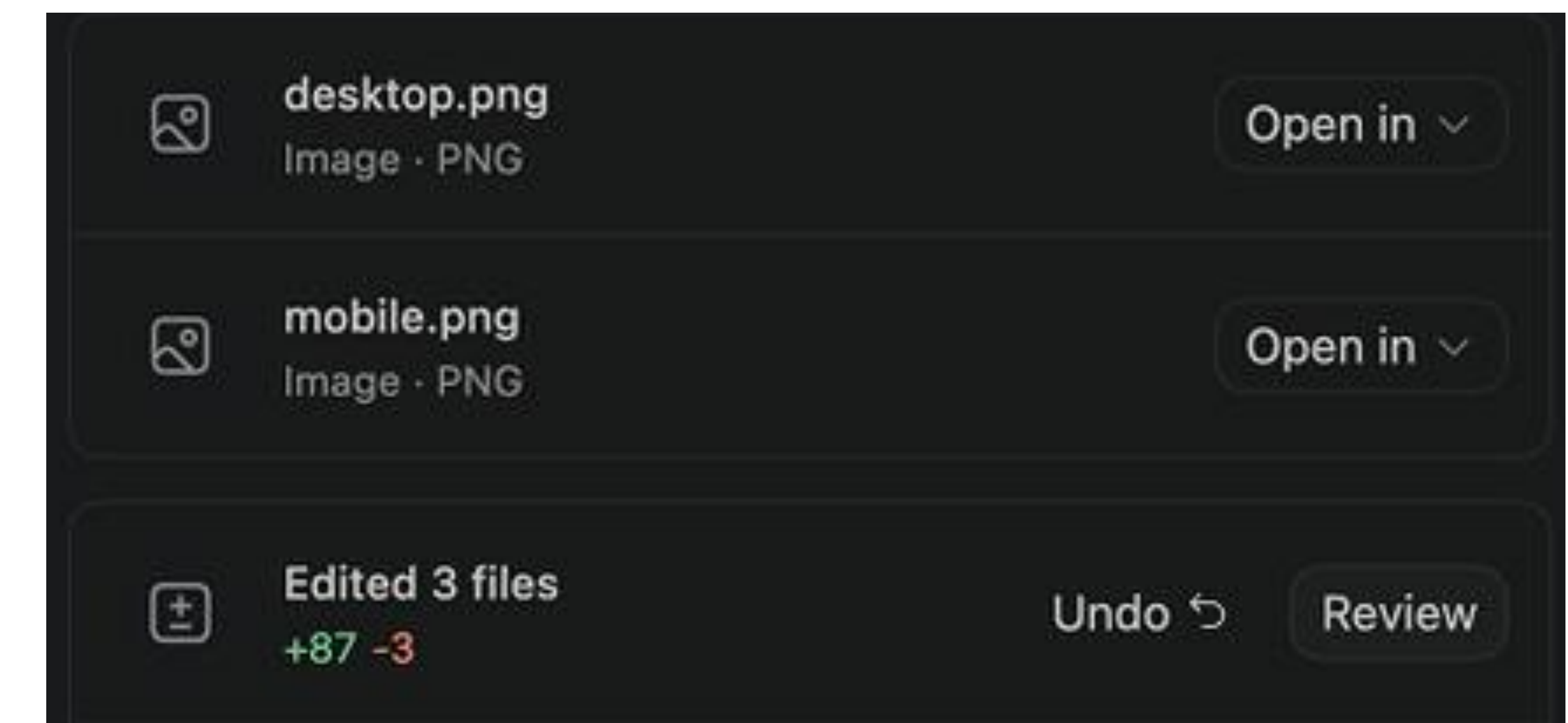
Codex: Review the Change and the Evidence

Look for the diff, result files, and review controls

CHANGED LINES

```
src/components/react/illustrative/CodexSurfaceFeatureIllustration.react.tsx +1 -1 v
441 unmodified lines
442   return (
443     <div className="grid h-full place-items-center p-[6cqw] text-[#161616] dark:text-white">
443     <div className="grid h-full place-items-center p-[8cqw] text-[#161616] dark:text-white">
444     <div className="grid h-full w-full place-items-center rounded-[2.6cqw] border border-black/10 bg-white shadow-[0 16px 45px rgba(0,0,0,0.1)]">
```

RESULTS + REVIEW



Read the actual patch

[Codex ↗](#)

Open the results

[Claude Code: terminal route ↗](#)



Reuse It. Then Build Your Own.

k-skill → Skill Creator → a verified result

Two Labs, Different Questions

Reuse an existing procedure. Create a new one.

1 · k-skill

2 · Skill Creator

GeekNews search

Public-data API

Can I reuse this workflow?

Can I make it repeatable?

[Open the step-by-step practice page ↗](#)

Check the Workspace First

ChatGPT desktop Work · a fresh task folder

- 1 Skills** Find the actual Skill picker
- 2 Runtime** Node.js / npx + Python 3
- 3 Vercel** Sign in in your own browser

Web chat and local Skill setup are different.

k-skill: Familiar Tasks, Reusable Workflows

Explore a Skill for a task you already understand



Train timetables

HWP documents

Local information

Inspect the procedure, inputs, and required tools. ↗

Install One Skill, Not Everything

k-skill · geeknews-search

- 1 Read** Purpose, commands, dependencies
- 2 Install** Only the selected Skill folder
- 3 Find** Open it in the real Skill picker

A downloaded file is not an executed Skill.

Turn News into a Reading Brief

Select @geeknews-search · then inspect the result

List 5 recent items. Keep the source links.

Pick 2 worth reading and explain why.

Separate feed summaries from full articles.

Use a returned title keyword for the next search.

The current feed is not the whole web.

Open the Source Yourself

Returned title → original page

▲ Real-SWE: 비공개 실무 기업 코드베이스에서 AI 모델을 평가하는 벤치마크 [withspecific.com](#)

1P by GN+ 3시간전 | ★ favorite | 댓글 1개

- Real-SWE는 실제 기업에서 사용 허가를 받은 비공개 프로덕션 코드베이스와 업무를 바탕으로, 코딩 에이전트가 소프트웨어 엔지니어의 실무를 수행할 수 있는지 평가함
- 모델 단독이 아니라 모델과 기본 하네스의 조합을 평가하며, 과제마다 8회 독립 실행한 평균 해결률은 Fable 5.10이 38.8%로 가장 높고 GPT-6 Astra가 33.8%로 뒤를 이음
- 과제는 짧은 지시에서 기업별 업무 규칙과 기존 구현을 찾아내는 능력을 요구함. 지시문 길이 중앙값은 1,742자지만 참조 솔루션이 수정한 파일 수 중앙값은 11개임
- 분석한 10개 과제 중 6개의 해결률이 15% 미만이며, 모든 과제를 해결한 모델은 없음. 가장 흔한 실패는 요구사항 누락임
- 10분 미만 실행도 71.4%가 실패해 짧은 실행에서도 어려움이 나타남. 실행 1회당 추정 비용은 모델에 따라 2.50~6.96달러임

Check the title, scope, and what the summary leaves out.



Build a Skill from Public Data

A real API → a reusable procedure

Find the Dataset Before Writing Code

Public Data Portal → Korea Heritage Service

DATA 공공데이터포털
.GO.KR

공공데이터 ▾

데이터활용 ▾

기업공공데이터 문제해결지원센터

정보공유 ▾

이용안내 ▾

XML



국가유산청_전국 지정문화재 현황

The catalog describes the data. The provider serves it.

Read the API as a Contract

Where to ask, what to send, what comes back

- 1 Endpoint** List and detail are different requests
- 2 Parameters** Keyword · page · result limit
- 3 Response** XML fields, IDs, and missing values

A successful HTTP response still needs validation.

List First. Then Request Details.

Never invent the detail identifiers

```
ccbaMnm1 = 경복궁  
pageUnit = 3  
pageIndex = 1
```

List → item IDs

```
ccbaKdcd  
ccbaAsno  
ccbaCtcd
```

IDs → detail

Keep IDs as strings—including leading zeros.

One Request, Three Different Counts

Actual check: Gyeongbokgung

11

Total matches

All matching records

3

Fetches items

The requested page

1

Visible item

After a local name filter

Counts can change. Check their meaning—not a memorized number.

Skill Creator Builds the Procedure

Select @skill-creator

- 1 Input** Heritage keyword + result limit
- 2 Output** Verified JSON + sources + timestamp
- 3 Boundary** No invented hours, fees, or deployment

heritage-brief is new; it is not a k-skill import.

Three Parts, Three Jobs

Read the generated files before invoking the Skill

SKILL.md

When + how

Keyword → facts →
checks

scripts/

**Repeatable
execution**

HTTP → XML → JSON

references/

API details

Endpoints + field
meanings

Run It on Input A

Select @heritage-brief · keyword: 경복궁

- 1 Call** Fetch the list and matching details
- 2 Save** heritage-a.json
- 3 Compare** One item against its official response

Inspect the actual call and saved data.

Same Skill, New Input

Keep A. Fetch B. Do not copy the old answer.

INPUT A

INPUT B

경복궁

첨성대

3 items returned

1 item returned

A limit of 3 does not mean “invent 3 results.”

Empty Is Not Broken

Test the boundary, not only the happy path

- 1 No matches** A valid response with zero items
- 2 Request failed** No verified answer from the source
- 3 Missing field** Keep it unknown; do not fill it in

Use a nonexistent keyword and a simulated network error.



Make the Result Useful

Verified data → a small explorable page

From API Facts to a Useful Page

A real response snapshot—not a made-up data set

입력 A · 경복궁

입력 B · 첨성대

현재 결과에서 이름 검색

전체 일치 11건 · 수집 3건 · 현재 표시 3건 | 조회: 2026. 9. 13. 오후 2:20:46

국보 · 서울특별시

경복궁 근정전

서울특별시 종로구 사직로 161 (세종로, 경복궁)

▶ 공식 설명 읽기

공식 API 응답 보기

국보 · 서울특별시

경복궁 경회루

서울특별시 종로구 사직로 161 (세종로, 경복궁)

▶ 공식 설명 읽기

공식 API 응답 보기

보물 · 서울특별시

경복궁 자경전

서울특별시 종로구 사직로 161 (세종로, 경복궁)

▶ 공식 설명 읽기

공식 API 응답 보기

Switch inputs. Search names. Open explanations and sources.

A Snapshot Is Not a Live API

Keep the architecture small and honest

- 1 Fetch** The agent runs the API helper
- 2 Embed** Reviewed JSON goes into index.html
- 3 Serve** The browser displays that snapshot

Live refresh needs additional server-side design.

Type a Name. Check the Count.

Actual browser check · 경회루 → one visible item

입력 A · 경복궁

입력 B · 첨성대

경회루

전체 일치 11건 · 수집 3건 · 현재 표시 1건 | 조회: 2026. 9. 13. 오후 2:20:46

국보 · 서울특별시

경복궁 경회루

서울특별시 종로구 사직로 161 (세종로, 경복궁)

▶ 공식 설명 읽기

공식 API 응답

Clear the query. Switch to B. Check that stale results disappear.

Verify More Than a Screenshot

A focused check is a reusable habit

The screenshot shows a web application interface. At the top, there are two buttons: '입력 A · 경북궁' (Input A · Gyeongbukgung) and '입력 B · 첨성대' (Input B · Cheonseongdae). Below these is a search bar with the placeholder text '현재 결과에서 이름 검색' (Search name in current results). Under the search bar, it says '전체 일치 1건 · 수집 1건 · 현재 표시 1건 | 조회: 2026. 9. 13. 오후 2:20:47' (Total 1 match · 1 collected · 1 currently displayed | Search: 2026. 9. 13. PM 2:20:47). The main content area shows '국보 · 경상북도' (National Treasure · Gyeongbuk) and '경주 첨성대' (Gyeongju Cheonseongdae). Below this, it says '경북 경주시 인왕동 839-1번지' (Gyeongbuk Gyeongju-si Inwang-dong 839-1). There are two links: '▶ 공식 설명 읽기' (▶ Read official explanation) and '공식 API 응답 >' (Official API response >).

Open the explanation

Follow the source link

Try an empty result

Use a narrow screen + keyboard

Reproduce → fix one thing → repeat the same check.



Connect, Deploy, Verify

Vercel MCP → a real preview → one update

Skill, MCP, and Account Access

Different responsibilities

- 1 Skill** The repeatable procedure
- 2 MCP** Tool discovery, requests, and results
- 3 OAuth** Which account may be used

Writing a Skill does not grant service access.

Connect the Vercel MCP Server

ChatGPT desktop · Settings → MCP servers

```
https://mcp.vercel.com
```

1 · Add

Streamable HTTP

2 · Save

Restart the connection

3 · Authenticate

Vercel OAuth window

Check /mcp, then make one read-only call.

Read the Tool Contract Before Deploying

Official Vercel documentation—not a deployment receipt

deploy_to_vercel

Deploy files directly to a new Vercel project without a Git repository or the Vercel CLI. Provide the file tree and a deployment target. Vercel creates the project if needed, detects the framework, and starts the build.

Parameter	Type	Required	Default	Description
<code>target</code>	string	Yes	-	Deployment target: <code>preview</code> for a shareable non-production URL or <code>production</code> to deploy to production
<code>name</code>	string	Yes	-	Project name. Vercel creates the project if it does not already exist
<code>files</code>	array	Yes	-	File tree to deploy. Provide source files only; Vercel installs dependencies and builds the project

files + name + target + the intended team

Read First. Then Choose a Destination.

Do not guess IDs or reuse an unrelated project

- 1 Inspect** Available tools + deployment docs
- 2 Select** A returned team ID
- 3 Name** A new workshop project

General text is not proof that a tool was called.

Deploy the Reviewed Files

Use the target fields on the practice page

- 1 Files** Complete index.html + embedded data
- 2 Target** preview, in the intended team
- 3 Receipt** Deployment ID + URL + build state

Ready, accessible, and usable are separate checks.

One Focused Update

Keep the data and existing behavior

- 1 Change** Only the main heading
- 2 Review** Read the diff
- 3 Verify** New URL + the same interactions

Preserve the previous deployment record.

Use Evidence to Find the Next Action

Keep setup failures separate from product failures

- | | | |
|----------|--------------------|--------------------------------------|
| 1 | No tool | Check the host, connection, and auth |
| 2 | Build error | Read that deployment's logs |
| 3 | Wrong page | Check the URL, files, and access |

If blocked, keep the Skill, JSON, and working local page.

Open-Source Coding Workspaces

The interface is open; model usage may still cost money



[Cline](#) ↗

Editor + approvals

open|code

[OpenCode](#) ↗

Terminal / IDE / app

```
$ aider
```

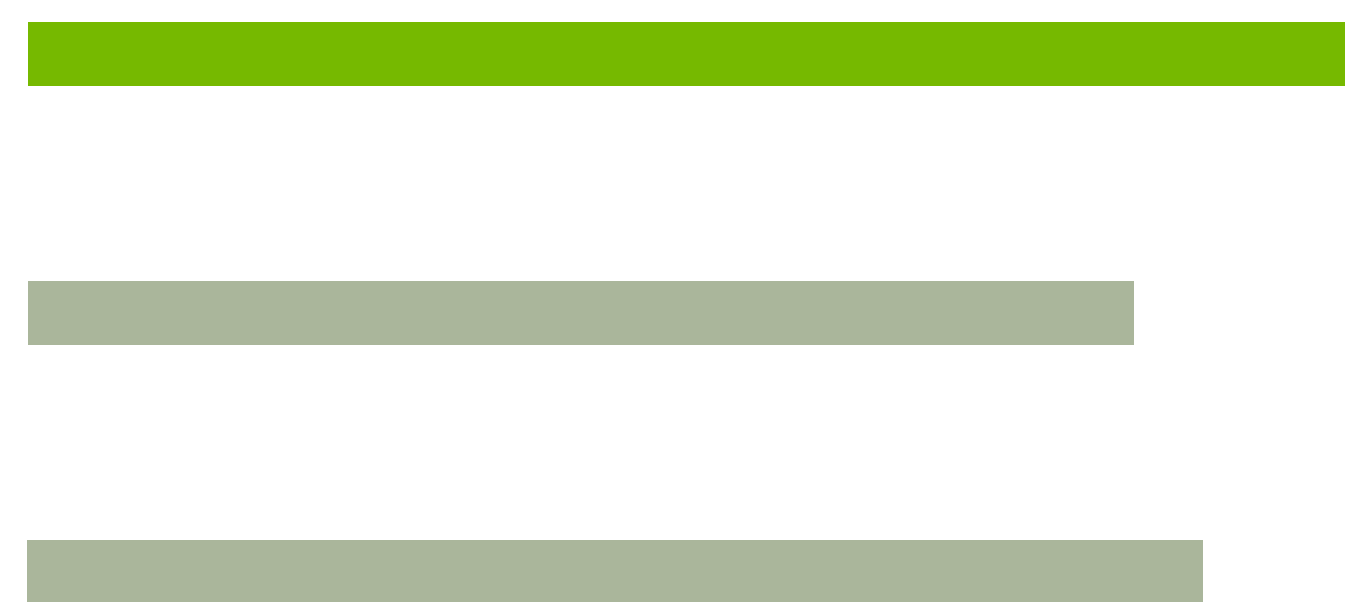
[Aider](#) ↗

Terminal + Git

Spec Kit: Start from a Clear Spec

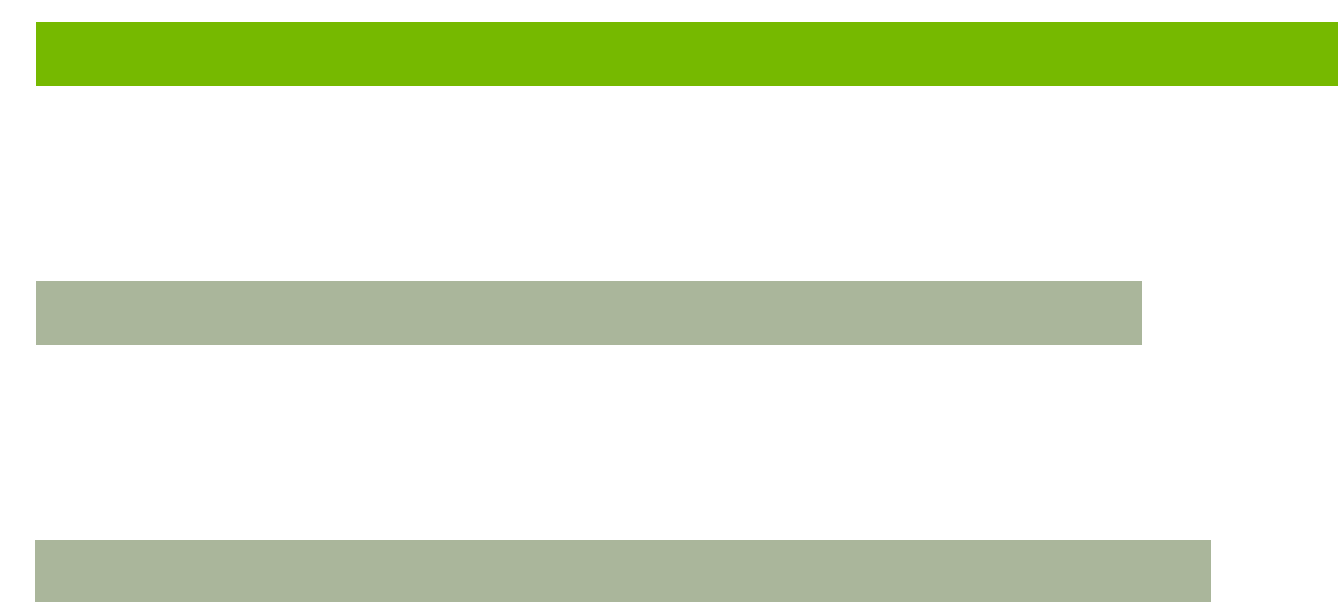
An optional workflow for a larger project

SPEC



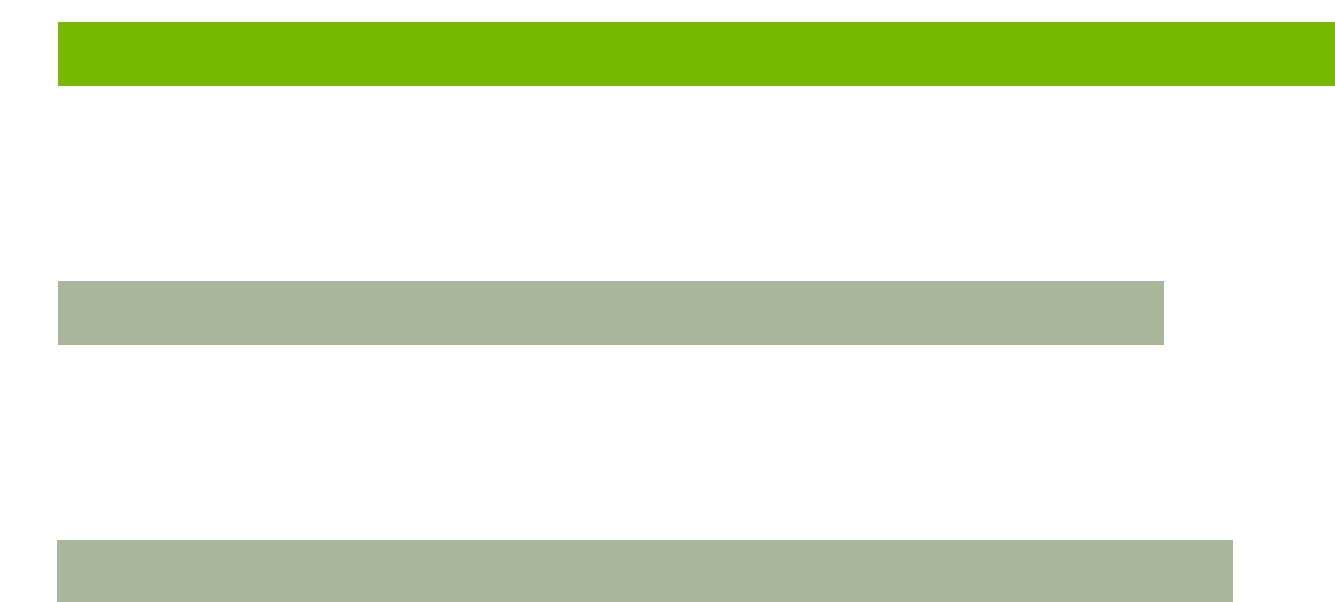
User + behavior

PLAN



Files + steps

TASKS



Small changes

[Spec Kit](#) ↗

Can Someone Else Reuse Your Work?

Pair review · use a new keyword

- 1 Skill** Explain the trigger and procedure
- 2 Evidence** Show a real response and source
- 3 Outcome** Demonstrate the page and one update

Take the pattern to a task you actually repeat.



Building RAG Agents with LLMs

Product Building with Agentic Coding

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.